



UNIVERSIDAD DE ANTIOQUIA
1803

Neutrino masses and dark matter

Scotogenic models

Diego Restrepo

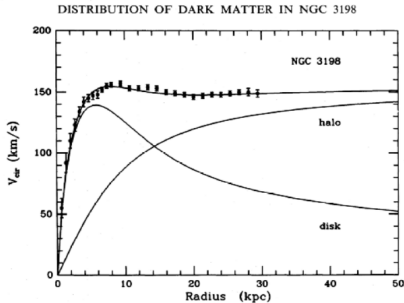
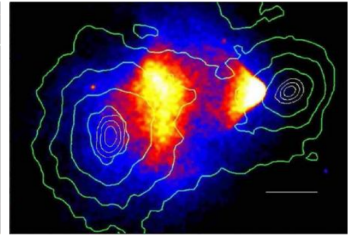
Instituto de Física
Universidad de Antioquia
Phenomenology Group
<http://gfif.udea.edu.co>



Evidences for Dark Matter

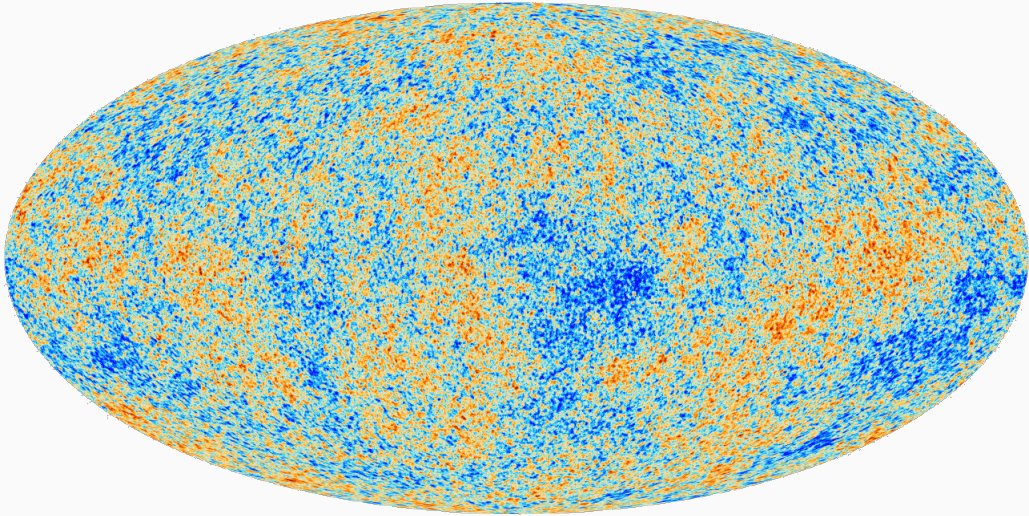
Several observations indicate the existence of non-luminous Dark Matter (*missing gravitational force*) at very different scales!

- * Galactic rotation curves
- * RC in Clusters of galaxies
- * Clusters of galaxies

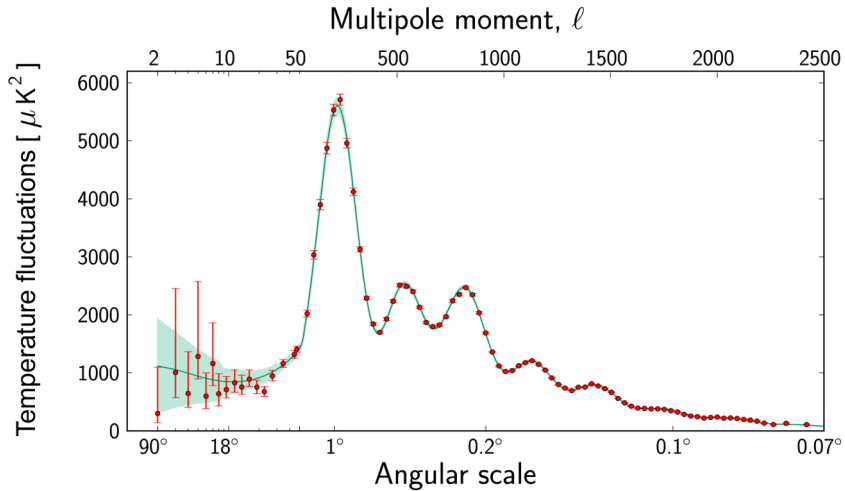


From Nicolas Bernal at NEMO-C (Medellín 2024)

Cosmic Microwave Background temperature: $T = 2.726 \text{ K}$ with $\Delta T/T \sim 10^{-6}$



The Cosmic Microwave Background - as seen by Planck. Credit: ESA and the Planck Collaboration

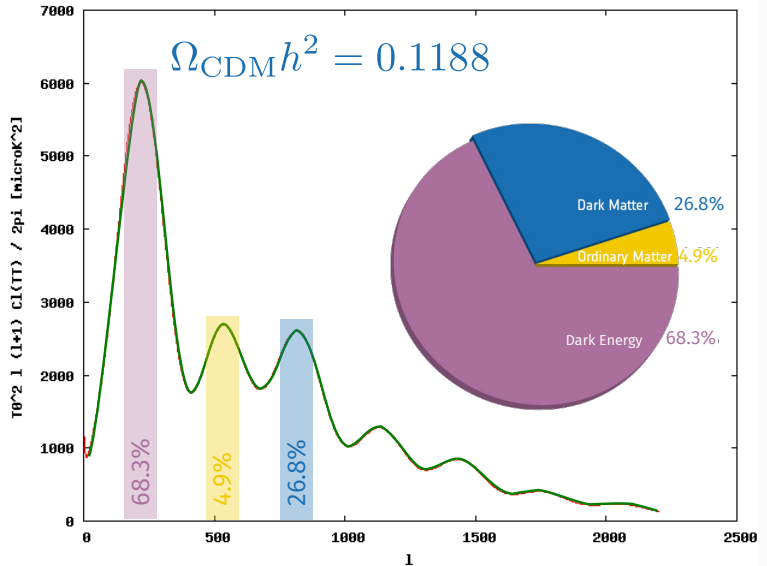


Planck's power spectrum of temperature fluctuations, ΔT , in the Cosmic Microwave Background. Credit: ESA and the Planck Collaboration

Λ CDM: $\Omega = 1$, $w = -1^\dagger$

Symbol	Value
$\Omega_b h^2$	0.022 30(14)
$\Omega_{\text{CDM}} h^2$	0.1188(10)
t_0	$13.799(21) \times 10^9$ year
n_s	0.9667(40)
Δ_R^2	2.441×10^{-9}
τ	0.066(12)

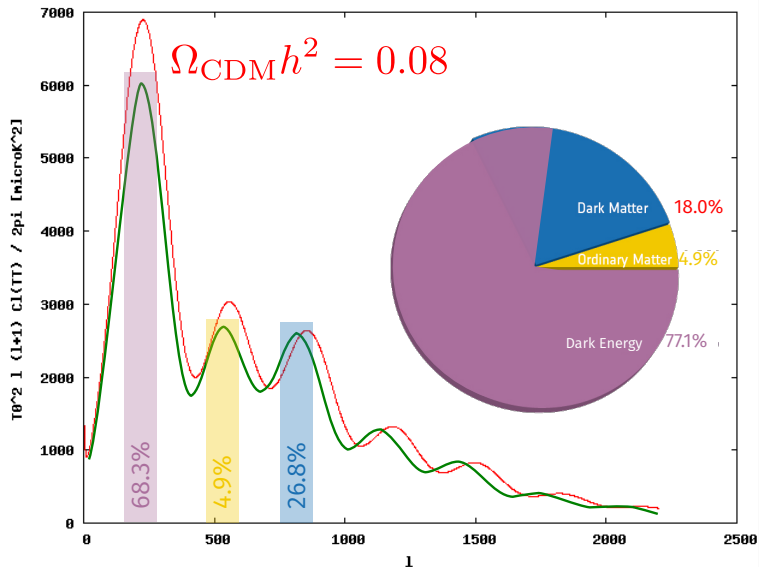
† Cosmological constant



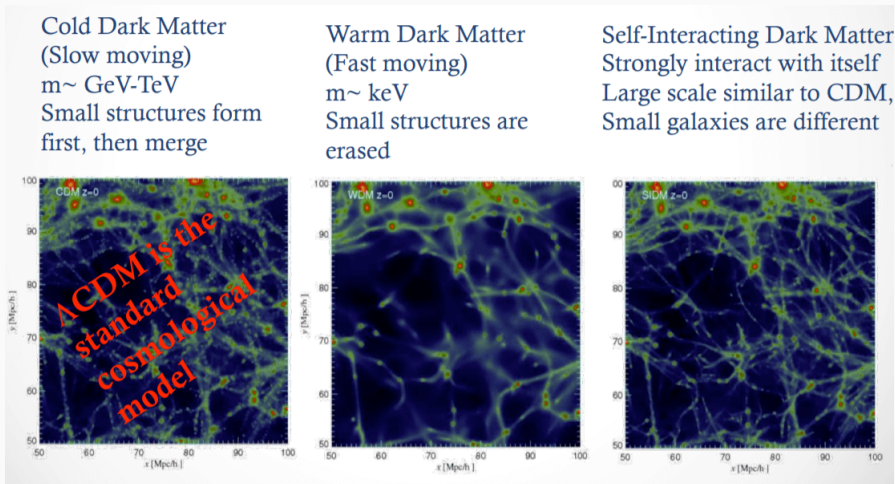
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Large scale structure simulations: Gas of not hot and almost collisionless dark matter **particles**



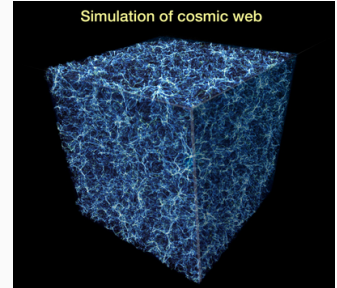
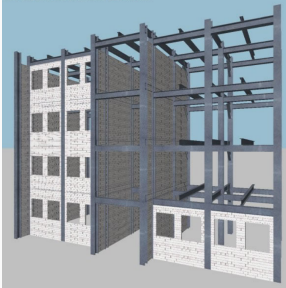
Credit: Arianna Di Cintio (Conference on Shedding Light on the Dark Universe with Extremely Large Telescopes, ICTP - 2018)

Particle: from elementary sub-eV to Primordial Black Hole of several solar masses

Cosmic web

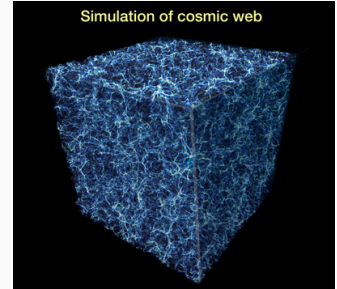
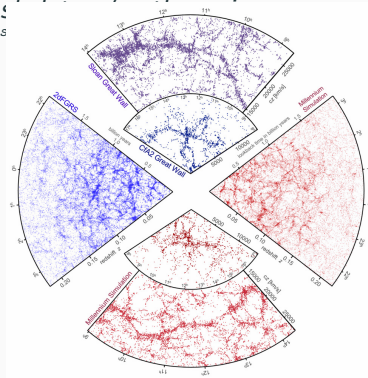
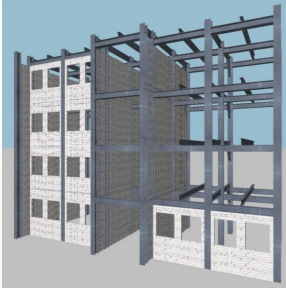
Dark matter connects clusters of galaxies with massive tendrils, forming a cosmic web that serves as an unseen skeleton for the universe.

<https://phys.org/news/2018-06-years-scientists-account-universe.html>



Cosmic web

Dark matter connects clusters of galaxies with massive tendrils, forming a cosmic web that serves as an unseen skeleton of the universe.
<https://phys.org/news/2018-06-years-s>

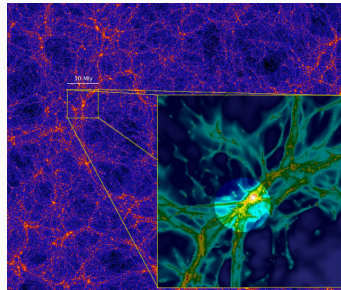
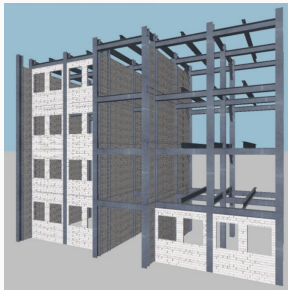


Galaxy redshift surveys vs large scale structure formation simulations: V. Springel, et al astro-ph/0604561 [Nature]

Cosmic web

Dark matter connects clusters of galaxies with massive tendrils, forming a cosmic web that serves as an unseen skeleton for the universe.

<https://phys.org/news/2018-06-years-scientists-account-universe.html>



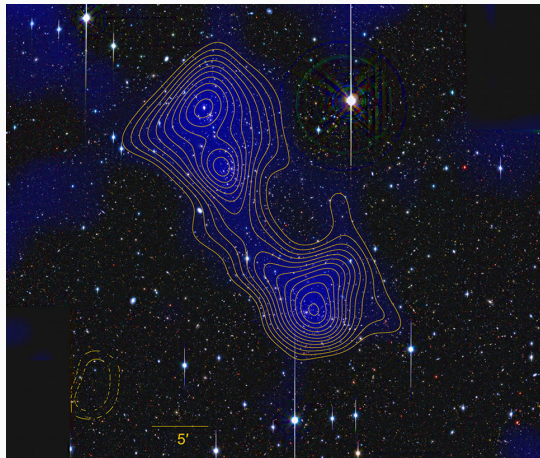
These great filaments are made largely of **dark matter** located in the space between galaxies and filled with 60% of the **primordial gas**!

An excess of a gas (20σ) is observed between Milky Way and Andromeda (M31): arXiv:1403.7528 [MNRAS] ¹

Clouds of HI likely embedded in a filament between M31 and M33: arXiv:1305.1631 [nature]

¹ See also: arXiv:1603.05400 [A&A]

A filament of dark matter between two clusters of galaxies



Supercluster system of three galaxy clusters

- Abell 222 (south) detected at $\sim 8\sigma$
- Abell 223 (north) double galaxy cluster seen at $\sim 7\sigma$

reconstructed surface mass density (blue)

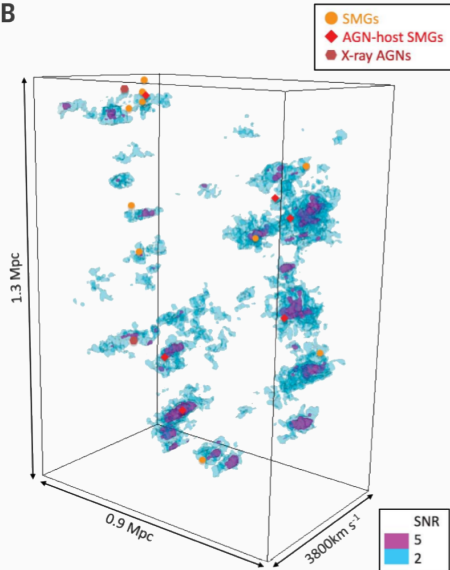
significance contours from 0.5σ to 2.5σ

J.P. Dietrich *et al*, arXiv:1207.0809 [Nature]

For a recent review see: arXiv:1905.08991

Three-dimensional pictures of Ly α filaments

B



The 3D distribution of Ly α filaments shown with

signal-to-noise ratio (SNR) > 5

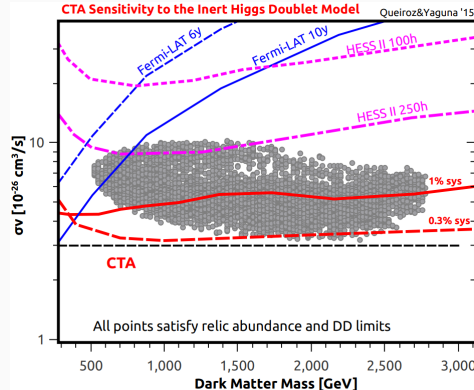
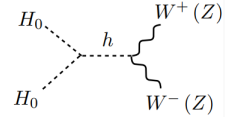
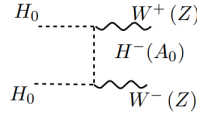
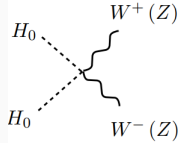
signal-to-noise ratio (SNR) > 2

H. Umehata *et al*, Science **366**, 97, 4 Oct 2019

SM + Inert Higgs doublet

$$\eta = \begin{pmatrix} \eta^+ \\ H_0 + iA_0 \end{pmatrix}$$

Lopez Honorez&et al'06



Baryon and Lepton Nonconserving Processes

#1

Steven Weinberg (Harvard U.) (1979)

Published in: *Phys.Rev.Lett.* 43 (1979) 1566-1570

[pdf](#) [DOI](#) [cite](#) [claim](#)

[reference search](#) [↻ 2,488 citations](#)

Naturally Light Dirac Neutrinos in Gauge Theories

#1

M. Roncadelli (Munich, Max Planck Inst.), D. Wyler (CERN) (Aug, 1983)

Published in: *Phys.Lett.B* 133 (1983) 325-329

[DOI](#) [cite](#) [claim](#)

[reference search](#) [↻ 116 citations](#)

Name	$SU(2)_L$	$U(1)_Y$	$U(1)_L$	Z_2
L	2	$-1/2$	-1	+
H	2	$1/2$	0	+

$$\mathcal{L}_M = \frac{y_M}{\Lambda} L \cdot H L \cdot H + \text{h.c} \rightarrow \Delta L = 2$$

Dark matter (radiative) realization $\rightarrow Z_2$

Effective neutrino masses

Majorana

Dirac

Baryon and Lepton Nonconserving Processes

Steven Weinberg (Harvard U.) (1979)

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Name	$SU(2)_L$	$U(1)_Y$	$U(1)_L$	Z'_2
L	2	$-1/2$	-1	+
H	2	$1/2$	0	+
ν_R	1	0	-1	-

$$\mathcal{L}_M = \frac{y_M}{\Lambda} L \cdot H L \cdot H + \text{h.c.} \rightarrow \Delta L = 2$$

Avoids Higgs mechanism for $\nu_R \rightarrow Z'_2$

$$\mathcal{L}_D = y_D \cancel{(\nu_R)^\dagger} L \cdot H + \text{h.c.}$$

$$\cancel{M_N \nu_R \nu_R} \rightarrow U(1)$$

Scotogenic one-loop realizations

Majorana

Dirac

Systematic study of the d=5 Weinberg operator at one-loop order

#1

Florian Bonnet (Wurzburg U.), Martin Hirsch (Valencia U., IFIC), Toshihiko Ota (Munich, Max Planck Inst.), Walter Winter (Wurzburg U.) (Apr, 2012)

Published in: *JHEP* 07 (2012) 153 • e-Print: [1204.5862](#) [hep-ph]

pdf DOI cite claim

reference search 197 citations

Models with radiative neutrino masses and viable dark matter candidates

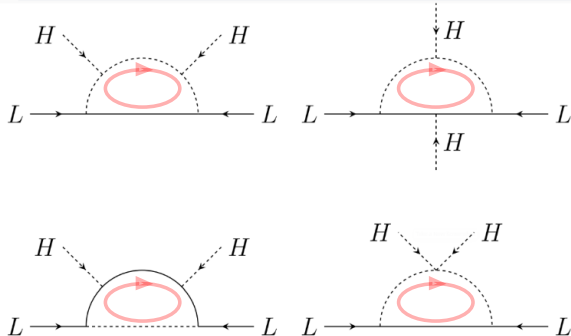
#1

Diego Restrepo (Antioquia U.), Oscar Zapata (Antioquia U.), Carlos E. Yaguna (Munster U., ITP) (Aug 16, 2013)

Published in: *JHEP* 11 (2013) 011 • e-Print: [1308.3655](#) [hep-ph]

pdf DOI cite claim

reference search 123 citations



Pathways to Naturally Small Dirac Neutrino Masses

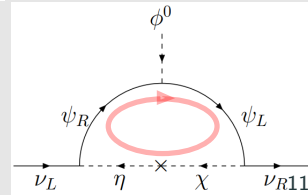
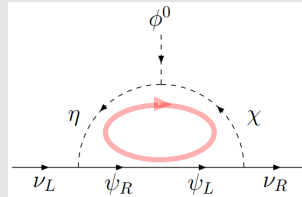
#1

Ernest Ma (UC, Riverside), Oleg Popov (UC, Riverside) (Sep 8, 2016)

Published in: *Phys.Lett.B* 764 (2017) 142-144 • e-Print: [1609.02538](#) [hep-ph]

pdf DOI cite claim

reference search 125 citations



Scotogenic one-loop realizations

Majorana

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Systematic study of the d=5 Weinberg operator at one-loop order

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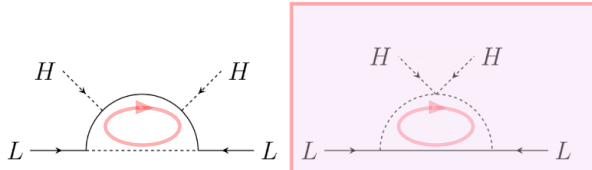
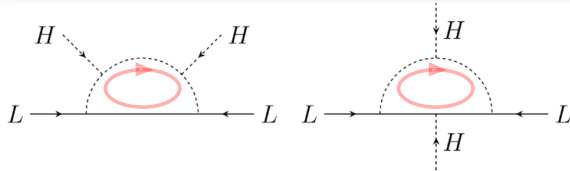
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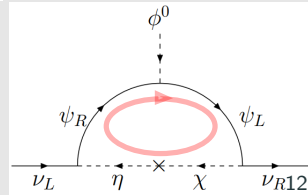
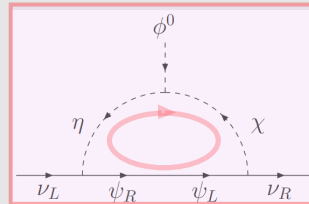
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Published in: *Phys.Lett.B* 764 (2017) 142-144 • e-Print: [1609.02538](#) [hep-ph]

[pdf](#) [DOI](#) [cite](#) [claim](#)

[reference search](#) [125 citations](#)

#1



Radiative seesaw mechanism at weak scale

#1

Zhi-jian Tao (Beijing, Inst. High Energy Phys.) (Mar, 1996)

Published in: *Phys.Rev.D* 54 (1996) 5693-5697 • e-Print: [hep-ph/9603309](https://arxiv.org/abs/hep-ph/9603309) [hep-ph]
[pdf](#)
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[34 citations](#)

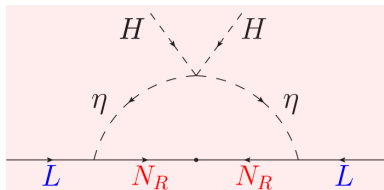
Verifiable radiative seesaw mechanism of neutrino mass and dark matter

#2

Ernest Ma (UC, Riverside) (Jan, 2006)

Published in: *Phys.Rev.D* 73 (2006) 077301 • e-Print: [hep-ph/0601225](https://arxiv.org/abs/hep-ph/0601225) [hep-ph]
[pdf](#)
[DOI](#)
[cite](#)
[claim](#)
[reference search](#)
[1,539 citations](#)

Name	$SU(2)_L$	$U(1)_Y$	Z_2
N_R	1	0	—
η	2	1/2	—



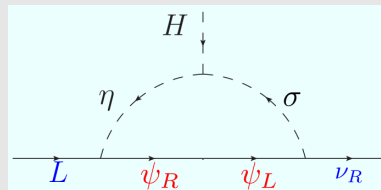
Radiative Neutrino Mass, Dark Matter and Leptogenesis

#1

Pei-Hong Gu (ICTP, Trieste), Utpal Sarkar (Ahmedabad, Phys. Res. Lab) (Dec, 2007)

Published in: *Phys.Rev.D* 77 (2008) 105031 • e-Print: [0712.2933](https://arxiv.org/abs/0712.2933) [hep-ph]
[pdf](#)
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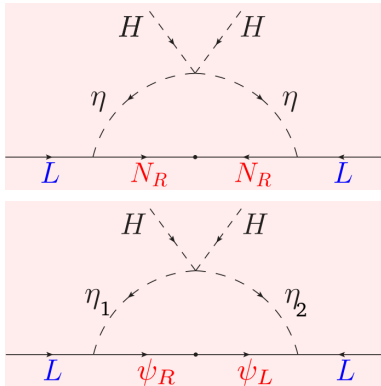
Name	$SU(2)_L$	$U(1)_Y$	Z_2	$U(1)_L$
ν_R	1	0	—	—1
ψ_R	1	0	+	$r-1$
ψ_L	1	0	+	$r-1$
η	2	1/2	+	r
σ	1	0	—	r



Radiative seesaw mechanism at weak scale

#1

Zhi-jian Tao (Beijing, Inst. High Energy Phys.) (Mar, 1996)

Published in: *Phys.Rev.D* 54 (1996) 5693-5697 • e-Print: [hep-ph/9603309](#) [hep-ph]
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New Scotogenic Model of Neutrino Mass with $U(1)_D$ Gauge Interaction

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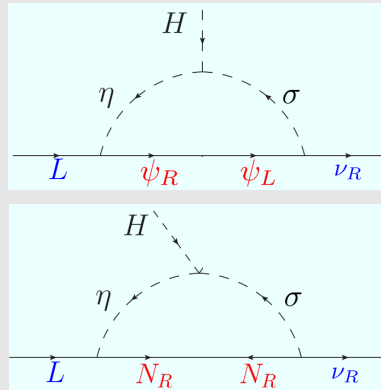
Ernest Ma (UC, Riverside), Ivica Picek (Zagreb U., Phys. Dept.), Branimir Radovčić (Zagreb U., Phys. Dept.) (Aug 24, 2013)

Published in: *Phys.Lett.B* 726 (2013) 744-746 • e-Print: [1308.5313](#) [hep-ph]
[pdf](#)
[DOI](#)
[cite](#)
[claim](#)
[reference search](#)
[86 citations](#)

Radiative Neutrino Mass, Dark Matter and Leptogenesis

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Published in: *Phys.Rev.D* 77 (2008) 105031 • e-Print: [0712.2933](#) [hep-ph]
[pdf](#)
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[reference search](#)
[124 citations](#)


Dirac neutrino mass generation from a Majorana messenger

#1

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U. and IIP, Brazil), Óscar Zapata (Antioquia U.) (Sep 20, 2019)

Published in: *Phys.Rev.D* 101 (2020) 3, 035004 • e-Print: [1909.09574](#) [hep-ph]
[pdf](#)
[links](#)
[DOI](#)
[cite](#)
[claim](#)
[reference search](#)
[15 citations](#)

Dark matter stability and Dirac neutrinos using only Standard Model symmetries

Cesar Bonilla (Munich, Tech. U.), Salvador Centelles-Chuliá (Valencia U., IFIC), Ricardo Cepedello (Valencia U., IFIC), Eduardo Peinado (Mexico U.), Rahul Srivastava (Valencia U., IFIC) (Dec 4, 2018)

Published in: *Phys.Rev.D* 101 (2020) 3, 033011 • e-Print: 1812.01599 [hep-ph]

Minimal radiative Dirac neutrino mass models

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U.), Carlos E. Yaguna (UPTC, Tunja), Óscar Zapata (Antioquia U.) (Dec 13, 2018)

Published in: *Phys.Rev.D* 99 (2019) 7, 075008 • e-Print: 1812.05523 [hep-ph]

pdf DOI cite claim reference search 46 citations

$$\mathcal{L}_5 = \frac{h}{\Lambda} (\nu_R)^\dagger L \cdot HS^* + \text{H.c.},$$

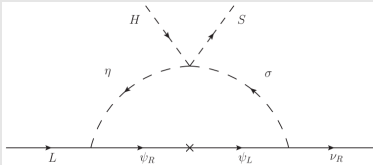


FIG. 6. T3-1-A-I ($\alpha = 0$): $B - L$ flux in the Dirac radiative seesaw.

	$(\nu_{Ri})^\dagger$	$(\nu_{Rj})^\dagger$	$(\nu_{Rk})^\dagger$						S
T3-1-A-I				$(\psi_R)^\dagger$	ψ_L	η_a	σ_a	—	
L	+4	+4	−5	r	$−r$	$1 − r$	$4 − r$	— 3	

From $Z_2 \times U(1)$ to a (secluded) $U(1)$ gauge symmetry with chiral fermions

In a fundamental theory the elementary fermions are expected to be chiral, i.e., their charges should not allow a mass term larger than the scale of spontaneous symmetry breaking. For any set of charges associated to a $U(1)$ gauge symmetry

$$\mathbf{Z} = [Z_1, Z_2, \dots, Z_N] ,$$

The triangle anomaly with three $U(1)$ gauge bosons on the external lines, i.e., the $[U(1)]^3$ anomaly, and with one $U(1)$ gauge boson and two gravitons on the external lines should also be cancelled

$$\sum_{\alpha=1}^N Z_{\alpha} = 0, \qquad \sum_{\alpha=1}^N Z_{\alpha}^3 = 0, \qquad (1)$$

$U(1)_Y$: no vector-like, i.e., pairs of fields with ‘equal-but-opposite’ charges

The hypercharge for one generation in the standard model can be seen as gauge Abelian symmetry $U(1)_Y$ with 15 left-handed Weyl massless fermions with integer charges

$$Y = [1, 1, 1, 1, 1, 1, -4, -4, -4, 2, 2, 2, -3, -3, 6]$$

which satisfy

$$\sum_i Y_i = 0, \quad \sum_i Y_i^3 = 0.$$

If we introduce an scalar, H , of charge 3 we can form Yukawa couplings through the Dirac pairs

$$u_\alpha = (1, -4), (1, -4), (1, -4), \quad d_\alpha = (1, 2), (1, 2), (1, 2), \quad e = (-3, 6)$$

and the *chiral fermions* acquire Dirac masses after the EWSB. They are degenerate because the additional color symmetry. One remains massless, $\nu_L = (-3)$. The remnant Z_3 symmetry guarantees the stability of the lightest quark

Secluded gauge $U(1)_D$ without vector-like fermions:

$$\mathbf{S} = [\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_{N'}]$$

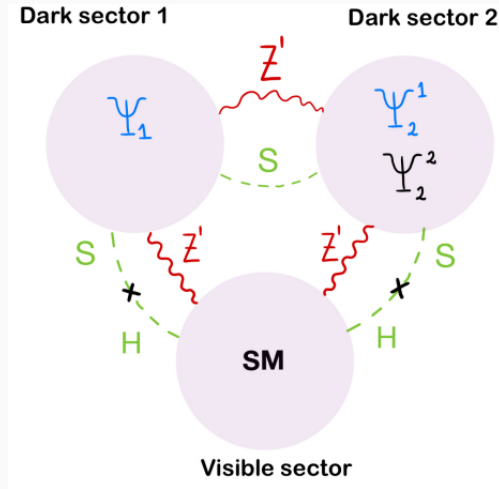
- *Dark Higgs mechanism*: Singlet dark Higgs ϕ acquires a vev and give mass to the dark photon

$$\mathcal{L} = i\psi_a^\dagger \overline{\sigma}^\mu (\partial_\mu - ig_D Z'_\mu) \psi_a - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{a < b} h_{ab} \psi_a \psi_b \phi^{(*)} + \text{h.c.} - V(\phi). \quad (2)$$

- S_α are the charges of SM-singlet right-handed chiral fermions with $N \geq 5$
 - χ_i *massless fermions* with $i = 1, \dots, N'$ with $N' \leq N$
 - ψ_a *multi-component dark matter*: massive after the spontaneous symmetry breaking of $U(1)_D$ with $a = N' + 1, \dots, N$
- *Larger parameter space*: Dark photon, Z' , exclusions instead of $B - L$ -like Z'
- Two mediators with Z' and ϕ masses are related by (arXiv:1610.03063)

$$h/g_D = \sqrt{2} m_\psi / m_{Z'}.$$

Multi-component and two-mediator DM with kinetic mixing



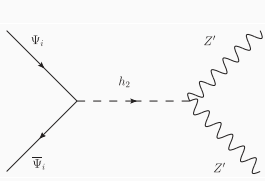
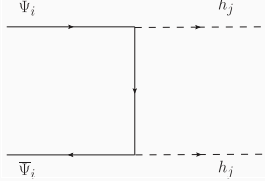
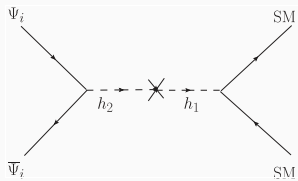
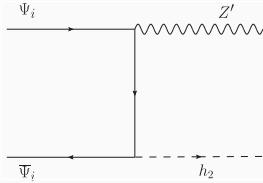
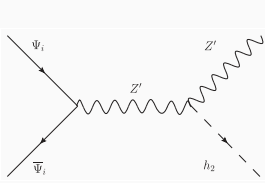
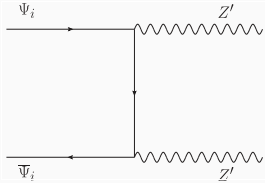
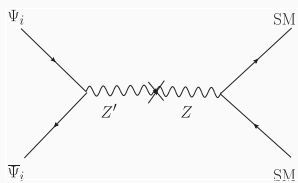
Two-mediator: Dark photon and dark Higgs

$$\begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \text{Re}(H^0) \\ \text{Re}(S) \end{pmatrix}$$

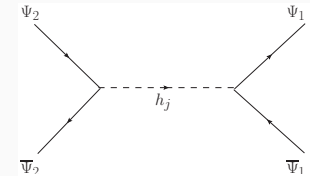
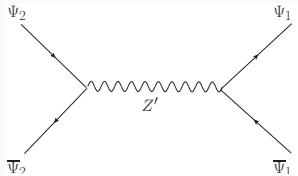
$$\mathcal{L}_{B'} \supset -\frac{\epsilon}{2} B'_{\mu\nu} B^{\mu\nu}$$

After the diagonalization of the kinetic terms

$$\mathcal{L}_{\text{mix}} \supset Z'_\mu (\epsilon e J_{EM}^\mu + g_D J_D^\mu)$$



DM conversion channels



Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

$$[\nu, \nu, (\nu), \psi_1, \psi_2, \dots, \psi_N]$$

$$\chi_1 \rightarrow \nu_{R1}, \dots, \chi_{N_\nu} \rightarrow \nu_{RN_\nu}, \quad 2 \leq N_\nu \leq 3,$$

$$\mathcal{L}_{\text{eff}} = h_{\nu}^{ij} (\nu_{Ri})^{\dagger} \epsilon_{ab} L_j^a H^b \left(\frac{\phi^*}{\Lambda} \right)^{\delta} + \text{H.c.}, \quad \text{with } i, j = 1, 2, 3,$$

ϕ is the complex singlet scalar responsible for the SSB of the anomaly-free gauge symmetry and give mass to all ψ_a

$$\phi = -\frac{\nu}{\delta},$$

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \rightarrow \nu_{R1}, \chi_2 \rightarrow \nu_{R2}, \quad N_\nu = 2,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{aj} (\nu_{Ra})^\dagger \epsilon_{bc} L_j^b H^c \left(\frac{\phi^*}{\Lambda} \right) + \text{H.c.}, \quad \text{with } j = 1, 2, 3,$$

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

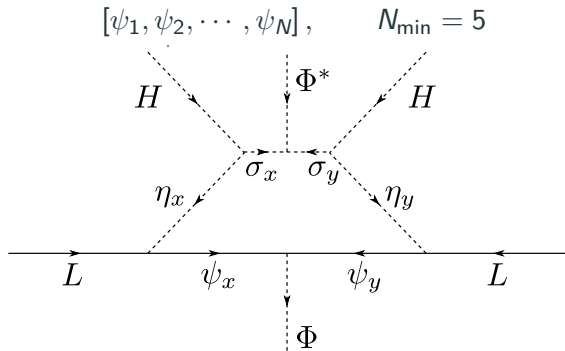
$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \rightarrow \nu_{R1}, \chi_2 \rightarrow \nu_{R2}, \quad N_\nu = 2,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{aj} (\nu_{Ra})^\dagger \epsilon_{bc} L_j^b H^c \left(\frac{\phi^*}{\Lambda} \right) + \text{H.c.}, \quad \text{with } j = 1, 2, 3,$$

$$z = [5, 5, -3, -2, 1, -6] \rightarrow \phi = -5 \rightarrow [(5, 5), (-3, -2), (1, -6)]$$

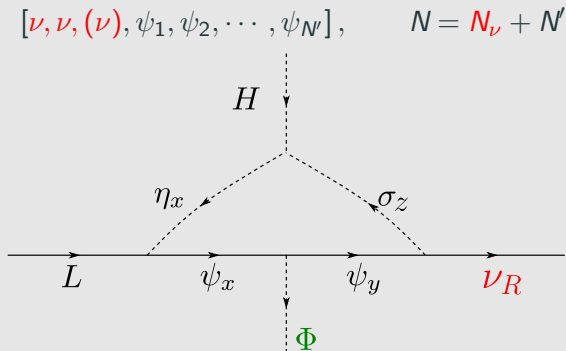
$$\mathcal{L} \subset h_{(-3, -2)} \psi_{-3} \psi_{-2} \phi^* + h_{(1, -5)} \psi_1 \psi_{-6} \phi^* + \text{h.c.}$$



$$\frac{y}{\Lambda} \bar{L} \bar{L} H H \rightarrow \frac{y}{\Lambda} \bar{L} \bar{L} H H \frac{\phi}{\Lambda} \frac{\phi^*}{\Lambda}$$

ϕ give mass to all ψ_a

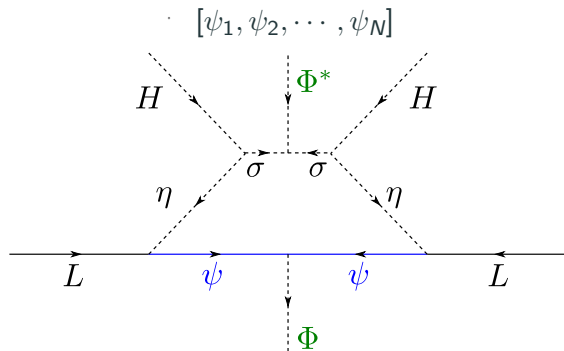
$\sim 1\,000$ solutions $[x, y, \dots]$ with $N \geq 8$



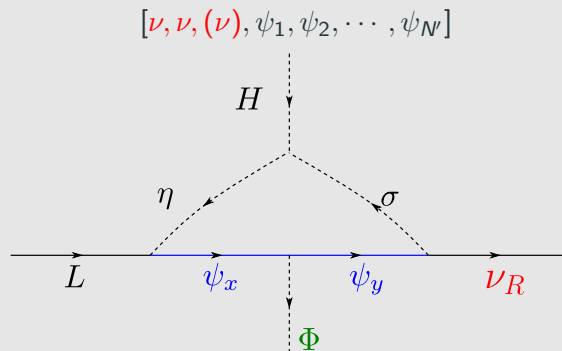
$$y(\nu_R)^\dagger \bar{L} H \rightarrow y(\nu_R)^\dagger \bar{L} H \frac{\phi}{\Lambda}$$

ϕ give mass to all ψ_a

$\sim 1\,000$ solutions $[\nu, \nu, (\nu), x, y, \dots]$ with $N \geq 6$



$$\phi = 2 \rightarrow [\underbrace{1, 1}_{\psi_a}, (2, -4), (4, -6), (3, -5)]$$



$$\phi = 9 \rightarrow [9, 9, 9, \underbrace{(1, -10), (1, -10)}_{\psi_a}, (-4, -5)],$$

Anomaly-free chiral $U(1)_D$ and its scotogenic implication

Chi-Fong Wong (Macau U. Sci. Tech., SSI) (Aug 19, 2020)

Published in: *Phys.Dark Univ.* 32 (2021) 100818 • e-Print: [2008.08573](https://arxiv.org/abs/2008.08573) [hep-ph]

#1

Anomaly-free Abelian gauge symmetries with Dirac scotogenic models

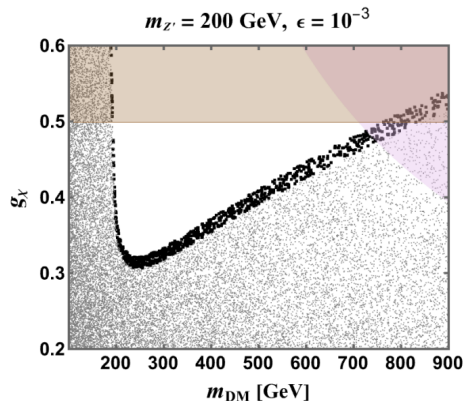
Nicolás Bernal (Antonio Narino U.), Julián Calle (Antioquia U.), Diego Restrepo (Antioquia U.) (Feb 11, 2021)

Published in: *Phys.Rev.D* 103 (2021) 9, 095032 • e-Print: [2102.06211](https://arxiv.org/abs/2102.06211) [hep-ph]

#8

$$\phi = 2 \rightarrow [1, 1, (2, -4), (4, -6), (3, -5)]$$

Secluded DM with Majorana fermion candidate



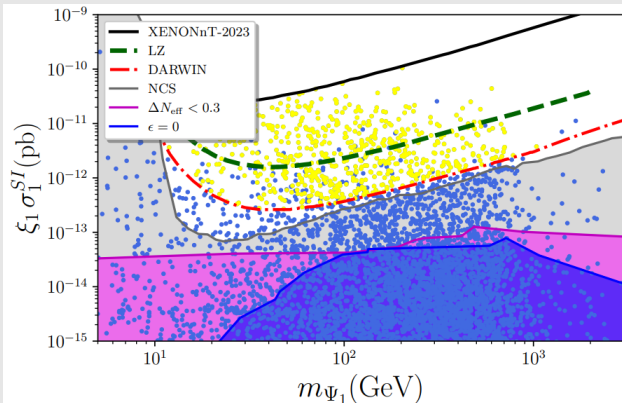
Chiral dark matter and radiative neutrino masses from gauged U(1) symmetry

K.S. Babu (Oklahoma State U.), Shreyashi Chakdar (Holy Cross Coll. and Munster U., ITP), Vishnu P. K (Sep 13, 2024)

e-Print: 2409.09008 [hep-ph]

$$\phi = 9 \rightarrow [9, 9, 9, (1, -10), (1, -10), (-4, -5)],$$

New decay channel: $Z' \rightarrow \bar{\nu}\nu$



In progress...

Many other venues

$$m_\nu \propto \frac{\langle H^0 \rangle^2}{\Lambda} \times \left(\frac{1}{16\pi^2} \right)^n \times \epsilon \times \left(\frac{\langle H^0 \rangle}{\Lambda} \right)^{d-5},$$

Other
representa-
tions

Other
Topologies

Two or
three
loops

Extra
coupling
suppression

Effective
operators
of higher
dimension

Scotogenic models, which explain dark matter and neutrino masses and mixings, have diverse realizations with a wide range of predictions through various portals and messengers.

When applied to chiral dark matter models arising from Abelian gauge symmetries, the stability of dark matter is automatically ensured, giving rise to a rich secluded dark sector.

Thanks!